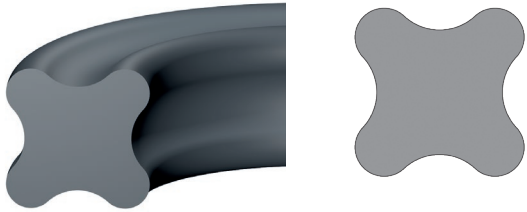


X-Rings XR



Description

X-rings are double-acting, circular sealing elements with an X-shaped square cross-section. X-rings are relatively similar to O-rings in terms of handling and use, but they offer some advantages in dynamic applications.

Special properties

- They exhibit no tendency to twist during reciprocating movements and installation thanks to the special four-lip profile
- Low friction
- Recess between the sealing lips serves as a lubricant reservoir, lower breakaway forces and static friction
- Unlike in O-rings, the flash is between the sealing lips and has no negative effect on the sealing behaviour
- Optimised sealing behaviour in dynamic applications thanks to the special distribution of the contact pressure over the profile

Applications

Versatile as a dynamic seal at low and medium speeds and as a static seal

- **Dynamic seal** for reciprocating movements on pistons, rods, etc.
- **Dynamic seal** for pivoting, rotating or helically moving shafts, spindles, etc.
- **Static seal**, radial or axial for flanges, covers, bushings, pipes, etc.

Materials

Standard materials

NBR 70 black, FKM 70 black

Special materials

EPDM, VMQ, FFKM, and others on request

Dimensions

Standard AS 568
and many additional sizes

Usage parameters

NBR 70 black

Temperature: -30°C to +100°C
Pressure: **Static** up to 5 MPa
With support ring up to 40 MPa
Dynamic
Reciprocating up to 5 MPa
With support ring up to 30 MPa
Rotating up to 5 MPa
With support ring up to 15 MPa
Speed: ≤ 0.5 m/s reciprocating
≤ 2 m/s rotating, for short periods
(Special design required)

Media / good resistance to:

- mineral oil-based oils and greases
- aliphatic hydrocarbons
- hydraulic oils H, H-L, H-LP
- hydraulic fluids HFA, HFB, HFC
- water up to 80°C

FKM 70 black

Temperature: -15°C to +200°C
Pressure: **Static** up to 5 MPa (50 bar)
With support ring up to 40 MPa
Dynamic
Reciprocating up to 5 MPa (50 bar)
With support ring up to 30 MPa
Rotating up to 5 MPa
With support ring up to 15 MPa
Speed: ≤ 0.5 m/s reciprocating
≤ 2 m/s rotating, for short periods
(Special design required)

Media / good resistance to:

- mineral oil-based oils and greases
- synthetic oils and greases
- aliphatic, aromatic and chlorinated hydrocarbons
- hydraulic fluids HFD
- fuels
- non-polar solvents
- ozone, weather, ageing, UV radiation
- water up to 80°C

Installation

Please follow our installation instructions